

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 (B)

$$f(\theta) = \min. (|2x - 7| + |x - 4| + |x - 2 - \sin \theta|)$$

$$\text{Let } g(x) = |2x - 7| + |x - 4| + |x - 2 - \sin \theta|$$

$$g(x)|_{\min} = g\left(\frac{7}{2}\right) = \frac{1}{2} + \frac{3}{2} - \sin \theta = 2 - \sin \theta = f(\theta)$$

$$\therefore f(\theta)|_{\max} = 3$$

Q.2 (B)

$$\cos^{-1}x + 2\cos^{-1}x + 3\cos^{-1}x = \pi + 2\pi + 3\pi$$

$$\Rightarrow x = -1.$$

Q.3 (C) 2

$$g(x) = 2\sqrt{2} \sin x + \cos x$$

$$g(x) \in [-3, 3]$$

$$y = \sqrt{g(x) - 2} + \sqrt{4 - g(x)}$$

$$y^2 = g(x) - 2 + 4 - g(x) + 2\sqrt{g(x) - 2}\sqrt{4 - g(x)}$$

$$y^2 = 2 + 2\sqrt{4g - g^2 - 8 + 2g} = 2 + 2\sqrt{-[g(x)]^2 + 6(g(x)) - 8} = 2 + 2\sqrt{1 - [g(x) - 3]^2}$$

$$\text{where } g(x) = 3 \text{ then } y^2_{\max} = 4$$

$$y_{\max} = 2 \text{ Ans.}$$

Q.4 (B)

Sol. $2x \in [-1, 1]$

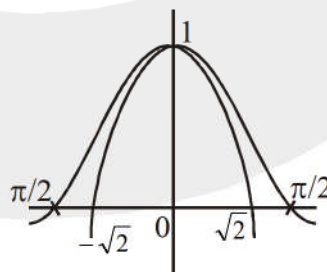
$$x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$$\text{and } \cos x + \frac{x^2}{2} - 1 > 0$$

$$\cos x > 1 - \frac{x^2}{2} \text{ from graph}$$

$$x \in \mathbb{R} - \{0\}$$

$$\text{Domain is } \left[-\frac{1}{2}, \frac{1}{2}\right] - \{0\}$$



Q.5 (A)

Sol. $f(x) = 2\tan^{-1}(2013^x) - \frac{\pi}{2}$

$$f(-x) = 2\tan^{-1}(2013^{-x}) - \frac{\pi}{2}$$

$$= 2\tan^{-1}\left(\frac{1}{2013^x}\right) - \frac{\pi}{2} = 2\cot^{-1}(2013^x) - \frac{\pi}{2} = 2\left[\frac{\pi}{2} - \tan^{-1}(2013^x)\right] - \frac{\pi}{2}$$

$$= \frac{\pi}{2} - 2\tan^{-1}(2013^x) = -\left[2\tan^{-1}(2013^x) - \frac{\pi}{2}\right] \Rightarrow f(-x) = -f(x) \Rightarrow f(x) \text{ is odd.}$$

$$\text{As, } f(-5) = -f(5) \Rightarrow f^2(-5) = f^2(5) \Rightarrow f^2(5) - f^2(-5) = 0 \text{ Ans.}$$

Q.6 (C)

Sol. $f(x) = \sqrt{6 - 5^{\sin^{-1}x} - 5^{1-\sin^{-1}x}}$ and $g(x) = \sqrt{5^{\sin^{-1}x}}$
 $x \in [-1, 1]$

$$6 - 5^{\sin^{-1}x} - 5^{1-\sin^{-1}x} = 6 - 5^{\sin^{-1}x} - \frac{5}{5^{\sin^{-1}x}} = -\left[\frac{(5^{\sin^{-1}x})^2 - 6(5^{\sin^{-1}x}) + 5}{5^{\sin^{-1}x}}\right]$$

$$= -\frac{(5^{\sin^{-1}x} - 5)(5^{\sin^{-1}x} - 1)}{5^{\sin^{-1}x}} = \frac{4 - (5^{\sin^{-1}x} - 3)^2}{5^{\sin^{-1}x}}$$

$$\Rightarrow y = h(x) = f(x) \cdot g(x) = \sqrt{-(5^{\sin^{-1}x} - 5)(5^{\sin^{-1}x} - 1)} = \sqrt{4 - (5^{\sin^{-1}x} - 3)^2}$$

$$\text{for domain } 1 \leq 5^{\sin^{-1}x} \leq 5 \Rightarrow \sin^{-1}x \in [0, 1]$$

$$\Rightarrow 0 \leq 4 - (5^{\sin^{-1}x} - 3)^2 \leq 4 \Rightarrow y \in [0, 2]$$

Q.7 (A)

$$f(x) = x^3 + px^2 + qx + 1$$

$$f(0) = 1 > 0; f(-1) = p - q < 0$$

$$\therefore x_0 \in (-1, 0)$$

$$2\tan^{-1}\left(\frac{1}{\sin x_0}\right) + \tan^{-1}\left(\frac{2\sin x_0}{1 - \sin^2 x_0}\right)$$

$$2\left(\tan^{-1}\left(\frac{1}{\sin x_0}\right) + \tan^{-1}(\sin x_0)\right)$$

$$2\left(-\frac{\pi}{2}\right) = -\pi \quad (\because \sin x_0 < 0)$$

Q.8 (B)

$$\therefore \sin^{-1}x + \cos^{-1}x^2 + \frac{\pi}{2} = 0$$

$$\Rightarrow \frac{\pi}{2} + \sin^{-1}x = -\cos^{-1}x^2$$



$$\begin{aligned} \Rightarrow \cos\left(\frac{\pi}{2} + \sin^{-1} x\right) &= \cos(-\cos^{-1} x^2) \\ \Rightarrow -x &= x^2 \\ \Rightarrow x &= 0, -1 \quad (x=0 \text{ rejected}) \\ \therefore x &= -1 = \alpha \end{aligned}$$

$$\therefore \text{Given expression} = \frac{\sec^{-1}(-1) - \tan^{-1}(-1)}{\cot^{-1}(-1) - \operatorname{cosec}^{-1}(-1)} = \frac{\pi - \left(\frac{-\pi}{4}\right)}{\frac{3\pi}{4} - \left(\frac{-\pi}{2}\right)} = \frac{\frac{5\pi}{4}}{\frac{5\pi}{4}} = 1$$

Q.9 (C)

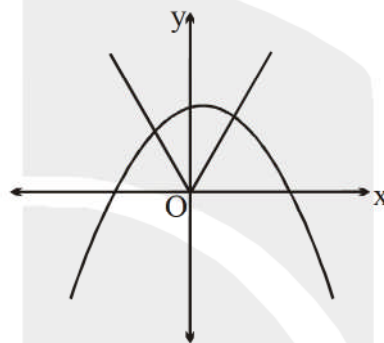
$$f(x) = \sum_{r=0}^{\infty} |x| \left(\tan^{-1} \left(\tan \frac{1}{2^r} \right) \right) = |x| \sum_{r=0}^{\infty} \frac{1}{2^r}$$

$$= |x| \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots \infty \right) = |x| \left(\frac{1}{1 - \frac{1}{2}} \right) = 2|x|$$

$$\therefore f(x) = 1 - x^2$$

$$\Rightarrow 2|x| = 1 - x^2$$

$\therefore$ Number of solutions = 2. **Ans.**



Q.10 (C)

$$(\sin^{-1} x) = (2 \tan^{-1} x)$$

Take sine on both sides, we get

$$x = \sin(2 \tan^{-1} x) \Rightarrow x = 0, \pm 1 \text{ **Ans.**}$$

Q.11 (C)

Q.12 (A)

Let GP is $a, ar, ar^2, \dots$ from given information

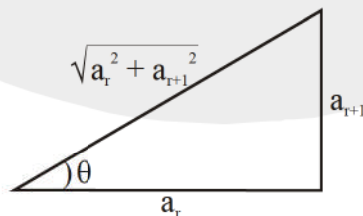
$$a = 2, r = 3$$

$$(i) \sum_{r=1}^{10} \tan \tan^{-1} \left(\frac{a_{r+1}}{a_r} \right)$$

$$\sum_{r=1}^{10} \frac{a_{r+1}}{a_r}$$

$$= \frac{a_2}{a_1} + \frac{a_3}{a_2} + \dots + \frac{a_{11}}{a_{10}}$$

$$= 10 \times r = 30.$$



$$(ii) \sin^{-1} \sin \left(\frac{ar}{a} \right) + \tan^{-1} \tan a$$

$$\sin^{-1} \sin 3 + \tan^{-1} \tan 2$$

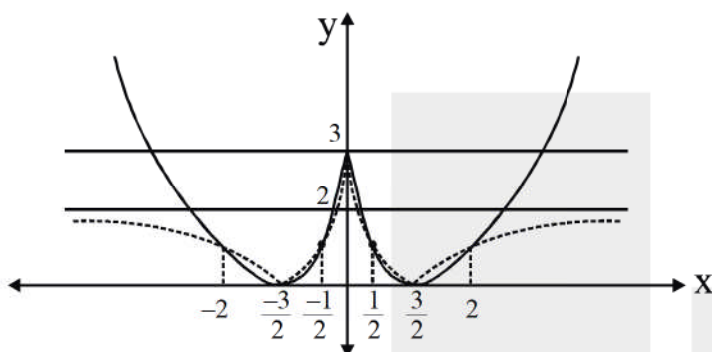
$$= \pi - 3 + 2 - \pi = -1. \text{ **Ans.**}$$



Q.13 (D)

Q.14 (B)

$$f(x) = \frac{|3 - 2|x||}{1 + |x|}$$



(i) $x = \pm 2, \pm \frac{3}{2}, \pm \frac{1}{2}, 0$

$$\therefore \sum_{i=1}^5 |x_i| = 7$$

(ii) $k(x) = p$ has four solutions.

$$\therefore p \in (0, 2)$$

Q.15 (AD)

$$f: \mathbb{R} - \{-1, k\} \rightarrow \mathbb{R} - \{\alpha, \beta\}$$

$$f(x) = \frac{(2x-1)(2x^2-4px+p^3)}{(x+1)(x^2-p^2x+p^2)}$$

For the domain to be $\mathbb{R} - \{-1, k\}$, k must be a repeated root of $x^2 - p^2x + p^2 = 0$

$$\therefore p^4 - 4p^2 = 0 \Rightarrow p^2(p^2 - 4) = 0, p = 0, \pm 2 \quad (p \geq 0)$$

For $p = 0$, $k = 0$

$$f(x) = \frac{(2x-1)2x^2}{(x+1)(x^2)}, \Rightarrow D_f: \mathbb{R} - \{-1, 0\}, R_f: \mathbb{R} - \{4, -2\}$$

$$\therefore \alpha + \beta = 2$$

For $p = 2$, $k = 2$

$$f(x) = \frac{(2x-1)2(x-2)^2}{(x+1)(x-2)^2} \Rightarrow D_f: \mathbb{R} - \{-1, 2\}, R_f: \mathbb{R} - \{4, 2\}$$

$$\alpha + \beta = 6. \text{ Ans.}$$

Q.16 (ABC)

$$\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x^2 - 1} + \tan^{-1} \tan y = k$$

$$0 \leq x \leq 1 \text{ and } 0 \leq x^2 - 1 \leq 1$$

Hence, $x = 1$

$$\frac{\pi}{2} + \frac{\pi}{2} + \tan^{-1} \tan y = k$$

$$\frac{-\pi}{2} < \tan^{-1} \tan y < \frac{\pi}{2}$$

$$\therefore k \in \left(\frac{\pi}{2}, \frac{3\pi}{2} \right)$$

Hence, $(p + q) = 4$.

Q.17 5

$$\because |x + y| \leq |x| + |y|$$

$$\text{But } |f(x) + (x^2 + 1)| \geq |f(x)| + |x^2 + 1|$$

$$\therefore |f(x) + (x^2 + 1)| = |f(x)| + |x^2 + 1| \text{ will be true iff } f(x) \cdot (x^2 + 1) \geq 0$$

$$\Rightarrow f(x) \geq 0$$

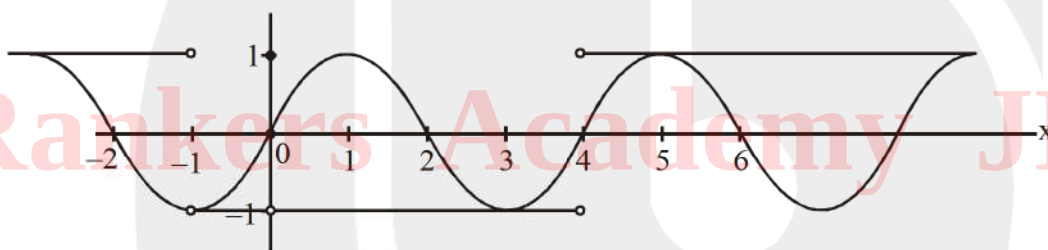
$$\text{But } f(x) \leq 0, \therefore f(x) = 0$$

$$\therefore \sum_{r=1}^5 (1 + f(r)) = \sum_{r=1}^5 1 = 5 \text{ Ans.}$$

Q.18 0041

$$f(x) = (x^2 - 3x - 4) = (x - 4)(x + 1)$$

$$g(f(x)) = \operatorname{sgn}((x - 4)(x + 1)), y = \sin \frac{\pi x}{2}$$



solutions are 3, 5, 9, 13, 17, -7, -3, 4

sum is 41.

Q.19 0055

$$10 - x \geq 0 \Rightarrow x \leq 10 ; \quad x \geq 0$$

$$x \in [0, 10]$$

$$\text{and } \frac{1}{3} \leq \frac{1}{2 - \sin 3x} \leq 1$$

i.e., always positive

$$\frac{1}{2 - \sin 3x} \neq 1, \text{ when } \sin 3x \neq 1$$

$$3x \neq n\pi + (-1)^n \frac{\pi}{2}$$

$$x \neq \frac{n\pi}{3} + (-1)^n \frac{\pi}{6}$$

$$\text{We get } x \neq \frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{21\pi}{6}, \dots$$

$$\text{but } \frac{21\pi}{6} > 10$$

$$\Rightarrow L = \frac{\pi}{6} + \frac{5\pi}{6} + \frac{9\pi}{6} + \frac{13\pi}{6} + \frac{17\pi}{6} =$$

$$L = \frac{45\pi}{6} \Rightarrow \frac{6L}{\pi} = 45$$

Q.20 2

$$\begin{aligned} & \sum_{n=0}^{\infty} \tan^{-1} \left(\frac{\tan^{-1} \left(\frac{(n+2) - (n+1)}{1 + (n+1)(n+2)} \right)}{1 + \tan^{-1} \left(\frac{1}{n+1} \right) \cdot \tan^{-1} \left(\frac{1}{n+2} \right)} \right) \\ &= \sum_{n=0}^{\infty} \tan^{-1} \left(\frac{\tan^{-1} \left(\frac{1}{n+1} \right) - \tan^{-1} \left(\frac{1}{n+2} \right)}{1 + \tan^{-1} \left(\frac{1}{n+1} \right) \cdot \tan^{-1} \left(\frac{1}{n+2} \right)} \right) \\ &= \sum_{n=0}^{\infty} \left(\tan^{-1} \left(\tan^{-1} \left(\frac{1}{n+1} \right) \right) - \tan^{-1} \left(\tan^{-1} \left(\frac{1}{n+2} \right) \right) \right) \\ &= \tan^{-1} \left(\tan^{-1} (1) \right) = \tan^{-1} \left(\frac{\pi}{4} \right) \end{aligned}$$

$$\therefore p = \pi$$

$$\left[\cos^{-1}(\cos(\pi-1)) \right] = [\pi-1] = 2 \text{ Ans.}$$